

# Fine TIME: Task-Specific Foundation-Model Adaptation

## 1 Question

Fine TIME evaluates whether task-specific adaptation of Chronos-2, Chronos-Bolt, and TS-ICL improves over the same frozen backbone on identical official TIME test windows. Each task is fitted only on its official training prefix. The initial protocol uses one seed, 100 full-model optimizer steps, no covariates, and the 90-task small profile.

## 2 Inherited frozen baselines

The executable surface contains a foundation benchmark of Chronos-2, Chronos-Bolt, TS-ICL, and Seasonal Naive; TimesFM-3 is excluded from the active experiment set. It also contains a Chronos-2 comparison between native multivariate targets, independent univariate targets, and other target histories supplied as past-only covariates. Seasonal Naive is deterministic and provides the taskwise MASE normalizer. It also fixes the evaluation grid to cells with finite target support, finite Seasonal Naive predictions on that support, and finite Seasonal Naive MASE. Every learned model uses this same grid; a non-finite forecast on expected support invalidates its task rather than changing the aggregation support. Inference timing excludes loading, dataset construction, metric computation, and artifact saving.

Two further executable comparisons use the same three learned backbones and evaluation grid. The maximum-context study evaluates five values per backbone, halving the model-specific maximum four times. Its summary aggregates taskwise MASE divided by the matched Seasonal Naive MASE for every model, context-size, and horizon-size cell. The normalization study compares the unchanged input with a per-window, per-variate z-score computed after context truncation. Constant inputs use unit scale, and every output quantile is transformed back to the target units before evaluation.

## 3 Fine-tuning comparison

Each model-task pair stores an independently adapted checkpoint, then evaluates the frozen and adapted states on the shared Seasonal-defined finite grid. The report requires exact pairs within a backbone, gives every task equal weight, and reports mean MASE, win rates, and paired PNG/PDF figures. Artifacts live below `outputs/<surface>/task_finetuning/`; reports and logs remain inside that experiment hierarchy.

## 4 Evidence boundary

Current result claims are limited to project-owned manifests and aggregates produced under the shared Seasonal evaluation grid. No Fine TIME run has yet completed, so the project makes no adaptation claim.